

Two-Sector DMP Model with Continuous Quality Sorting

Based on Mortensen-Pissarides (1994) and Lagos (2006)

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1 Environment

- **Time:** Continuous.
- **Sectors:** $j \in \{H, L\}$ denoting High-tech and Low-tech (non-high-tech). Sector-specific productivity factors are p_H and p_L .
- **Workers:** Unit mass of risk-neutral workers. Workers are heterogeneous in ability a , which is distributed continuously according to a CDF $F(a)$ with density $f(a)$ on support $[a_{\min}, a_{\max}]$.
- **Labor Force:** The total mass of workers with ability a is $f(a)$. Workers discount the future at rate r . Unemployed workers with ability a receive a flow benefit proportional to their ability, $b \cdot a$.
- **Firms:** Large pool of risk-neutral potential firms. A firm can open a vacancy in sector j at a flow cost c_j .

2 Search and Matching

Workers direct their search to a specific sector based on expected utility, but match randomly with firms within that sector.

- **Sector Choice:** Let $\phi_j(a) \in [0, 1]$ be the probability/fraction of unemployed workers of ability a who choose to search in sector $j \in \{H, L\}$.
- **Sector Pools:** The mass of unemployed workers searching in sector j is $u_j = \int \phi_j(a)u(a)da$, where $u(a)$ is the density of unemployed workers of ability a .
- **Market Tightness:** Let v_j be the number of vacancies in sector j . Market tightness is $\theta_j = v_j/u_j$.
- **Contact Rates:** The matching function is $M(u_j, v_j)$. The worker job-finding rate is $\lambda(\theta_j) = M(u_j, v_j)/u_j$. The firm vacancy-filling rate is $q(\theta_j) = M(u_j, v_j)/v_j$.
- **Pool Composition:** Conditional on a firm in sector j meeting a worker, the density of the worker's ability is $\pi_j(a) = (\phi_j(a)u(a))/u_j$.

3 Production and Endogenous Separation

- **Match Quality:** Upon meeting, the firm and worker draw a match-specific idiosyncratic quality shock $x \sim U[0, 1]$. The uniform distribution has mean 0.5, density $g(x) = 1$, and CDF $G(x) = x$.
- **Output:** A producing match between a sector j firm and a worker of ability a with shock x yields flow output $y_j(a, x) = p_j ax$.
- **Shocks:** New idiosyncratic shocks x' arrive at Poisson rate δ , drawn from the same uniform distribution $U[0, 1]$.
- **Exogenous Separation:** Matches are destroyed for exogenous reasons at rate s .
- **Endogenous Separation:** A match is continued if and only if the joint surplus is non-negative, defining a reservation match quality $R_j(a)$. If $x < R_j(a)$, the match separates.

4 Value Functions

4.1 Workers

The value of unemployment for a worker of ability a searching in sector j is $U_j(a)$:

$$rU_j(a) = ba + \lambda(\theta_j) \int_{R_j(a)}^1 [W_j(a, x) - U_j(a)] dx \quad (1)$$

The value of employment $W_j(a, x)$ satisfies:

$$rW_j(a, x) = w_j(a, x) + \delta \int_{R_j(a)}^1 [W_j(a, x') - W_j(a, x)] dx' - (\delta R_j(a) + s)[W_j(a, x) - U_j(a)] \quad (2)$$

Optimal Sorting: Workers choose the sector that maximizes their value:

$$U(a) = \max\{U_H(a), U_L(a)\} \quad (3)$$

4.2 Firms

The value of a vacancy in sector j is V_j :

$$rV_j = -c_j + q(\theta_j) \int \pi_j(a) \int_{R_j(a)}^1 [J_j(a, x) - V_j] dx da \quad (4)$$

Free entry implies $V_j = 0$ for all active sectors.

The value of a filled job $J_j(a, x)$ satisfies:

$$rJ_j(a, x) = p_j ax - w_j(a, x) + \delta \int_{R_j(a)}^1 [J_j(a, x') - J_j(a, x)] dx' - (\delta R_j(a) + s)J_j(a, x) \quad (5)$$

5 Surplus and Equilibrium Conditions

Wages are determined by continuous Nash Bargaining with worker bargaining power β :

$$W_j(a, x) - U_j(a) = \beta S_j(a, x) \quad \text{and} \quad J_j(a, x) - V_j = (1 - \beta) S_j(a, x) \quad (6)$$

The total match surplus $S_j(a, x) = W_j(a, x) - U_j(a) + J_j(a, x) - V_j$ satisfies:

$$(r + s + \delta) S_j(a, x) = p_j a x - r U_j(a) + \delta \int_{R_j(a)}^1 S_j(a, x') dx' \quad (7)$$

5.1 Job Destruction (Reservation Quality)

The reservation quality $R_j(a)$ is defined by $S_j(a, R_j(a)) = 0$:

$$p_j a R_j(a) - r U_j(a) + \delta \int_{R_j(a)}^1 S_j(a, x') dx' = 0 \quad (8)$$

Subtracting (8) from (7), we can express the surplus linearly in x :

$$S_j(a, x) = \frac{p_j a (x - R_j(a))}{r + s + \delta} \quad (9)$$

Substituting (9) into (8) and evaluating the integral over the uniform distribution $x \sim U[0, 1]$ yields:

$$p_j a R_j(a) - r U_j(a) + \frac{\delta p_j a}{r + s + \delta} \frac{(1 - R_j(a))^2}{2} = 0 \quad (10)$$

5.2 Derivatives of the Surplus Function

Using (9), we can derive the partial derivatives of the surplus $S_j(a, x)$ with respect to match quality x and ability a .

Differentiating with respect to x :

$$\frac{\partial S_j(a, x)}{\partial x} = \frac{p_j a}{r + s + \delta} \quad (11)$$

Differentiating with respect to a :

$$\frac{\partial S_j(a, x)}{\partial a} = \frac{p_j (x - R_j(a)) - p_j a R_j'(a)}{r + s + \delta} \quad (12)$$

Note on proportionality: Using the Nash bargaining solution, the value of unemployment can be written as:

$$r U_j(a) = b a + \lambda(\theta_j) \beta \int_{R_j(a)}^1 \frac{p_j a (x - R_j(a))}{r + s + \delta} dx = b a + \lambda(\theta_j) \beta \frac{p_j a (1 - R_j(a))^2}{2(r + s + \delta)} \quad (13)$$

Equating (13) with the expression for $r U_j(a)$ from (10) shows that a cancels out entirely from the equation determining $R_j(a)$. Therefore, $R_j(a) = R_j$ is independent of ability a , which implies $R_j'(a) = 0$.

Consequently, the partial derivative with respect to a from (12) simplifies to:

$$\frac{\partial S_j(a, x)}{\partial a} = \frac{p_j (x - R_j)}{r + s + \delta} \quad (14)$$

5.3 Job Creation (Free Entry)

Imposing $V_j = 0$ yields the job creation condition for sector j :

$$\frac{c_j}{q(\theta_j)} = (1 - \beta) \int \pi_j(a) \int_{R_j(a)}^1 S_j(a, x) dx da = (1 - \beta) \int \pi_j(a) \frac{p_j a (1 - R_j(a))^2}{2(r + s + \delta)} da \quad (15)$$

5.4 Steady-State Flow Equations

Let $e_j(a)$ be the density of employed workers of ability a in sector j .

$$u(a) + e_H(a) + e_L(a) = f(a) \quad (16)$$

The flow into employment in sector j must equal the flow out:

$$\lambda(\theta_j) \phi_j(a) u(a) (1 - R_j(a)) = e_j(a) (\delta R_j(a) + s) \quad (17)$$